Software Requirements Specification (SRS)

Dristi – AI-Based Assistance System for Visually Impaired People

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Dristi Project**, an AI-powered assistance system designed to help visually impaired individuals navigate safely, communicate efficiently, and share live location information with caregivers.

This document is intended for developers, project guides, testers, researchers, and future contributors involved in the development and maintenance of the system.

1.2 Scope

Dristi is an intelligent assistive platform that combines Artificial Intelligence, Computer Vision, IoT, GPS tracking, and voice-based interaction to support visually impaired users.

The system provides:

- Real-time obstacle detection
- Voice guidance and audio feedback
- Location sharing through Telegram bot integration
- Emergency alert support
- Caregiver monitoring support
- Navigation assistance
- Object recognition capabilities
- Raspberry Pi based deployment for portability

The project aims to increase independence, mobility, and safety for visually impaired individuals.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AI	Artificial Intelligence
CV	Computer Vision
GPS	Global Positioning System
IoT	Internet of Things

Term	Description
SRS	Software Requirements Specification
API	Application Programming Interface
Raspberry Pi	Single-board computer used for deployment
TTS	Text-to-Speech
STT	Speech-to-Text
YOLO	You Only Look Once object detection model

1.4 References

- IEEE SRS Documentation Standards
- Raspberry Pi Documentation
- OpenCV Documentation
- Telegram Bot API Documentation
- Python Official Documentation
- YOLO Object Detection Research Papers

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2. Overall Description

2.1 Product Perspective

Dristi is a standalone assistive system developed using embedded hardware and AI software modules. The system captures visual input through a camera connected to the Raspberry Pi, processes the information using Computer Vision models, and provides voice-based responses to the user.

The caregiver support system is integrated through Telegram APIs to provide real-time location tracking and emergency notifications.

2.2 Product Functions

The major functions of the Dristi system include:

1. Obstacle detection
2. Object recognition
3. Audio feedback generation
4. GPS-based live location sharing
5. Emergency SOS alert system
6. Telegram caregiver integration
7. Voice command support
8. Real-time environment analysis
9. Navigation assistance
10. User authentication and system monitoring

2.3 User Classes and Characteristics

User Type	Characteristics
Visually Impaired User	Primary user requiring navigation and assistance
Caregiver	Receives alerts and location updates
Administrator	Maintains system configurations
Developer	Updates and improves AI modules

2.4 Operating Environment

Hardware Requirements

- Raspberry Pi 4
- Camera Module / USB Camera
- GPS Module
- Speaker / Earphones
- Internet Connectivity
- Power Supply / Battery

Software Requirements

- Python 3.x
- OpenCV
- TensorFlow / PyTorch
- Telegram Bot API
- Flask / FastAPI (optional backend)
- Linux (Ubuntu/Raspberry Pi OS)

2.5 Design and Implementation Constraints

- Limited Raspberry Pi computational power
- Dependence on internet for Telegram communication
- Real-time processing constraints
- Battery limitations for portable usage
- Environmental lighting variations affecting detection accuracy

2.6 Assumptions and Dependencies

- Stable internet connection is available.
- GPS services are enabled.
- Camera hardware is functioning properly.
- AI models are pre-trained and optimized.
- Telegram services are accessible.

3. System Features and Functional Requirements

3.1 Obstacle Detection Module

Description

The system detects obstacles in front of the user using camera input and Computer Vision algorithms.

Functional Requirements

- The system shall continuously capture frames from the camera.
- The system shall identify nearby obstacles.
- The system shall estimate obstacle distance.
- The system shall generate audio warnings.
- The system shall operate in real time.

3.2 Object Recognition Module

Description

The module identifies common objects in the environment.

Functional Requirements

- The system shall recognize objects using AI models.
- The system shall announce object names through audio.
- The system shall support multiple object classes.
- The system shall update recognition results dynamically.

3.3 Voice Assistance Module

Description

Provides audio interaction between the system and the user.

Functional Requirements

- The system shall convert text to speech.
- The system shall support voice-based commands.
- The system shall provide navigation instructions.
- The system shall notify users about detected objects.

3.4 GPS Tracking and Location Sharing

Description

Allows live location tracking and sharing with caregivers.

Functional Requirements

- The system shall obtain GPS coordinates.
- The system shall send location updates periodically.
- The system shall integrate with Telegram bot APIs.
- The caregiver shall receive real-time location information.

3.5 Emergency SOS Module

Description

Enables emergency communication during unsafe situations.

Functional Requirements

- The system shall trigger SOS alerts.
- The system shall send emergency messages to caregivers.
- The system shall include live location in alerts.
- The system shall support manual and automatic emergency triggering.

3.6 Caregiver Monitoring System

Description

Provides monitoring capabilities for caregivers.

Functional Requirements

- Caregivers shall receive notifications.
- Caregivers shall track user location.
- Caregivers shall receive emergency alerts.
- Caregivers shall communicate through Telegram integration.

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4. External Interface Requirements

4.1 User Interface

The system provides:

- Voice-based interaction
- Telegram-based caregiver interface

• Command-line monitoring interface for administrators

4.2 Hardware Interfaces

Hardware	Purpose
Camera	Image capture
GPS Module	Location tracking
Speaker	Audio feedback
Raspberry Pi	Main processing unit

4.3 Software Interfaces

Software	Purpose
OpenCV	Image processing
TensorFlow/PyTorch	AI model execution
Telegram API	Communication and alerts
Python Libraries	Backend processing

4.4 Communication Interfaces

• Wi-Fi
• Internet APIs
• Telegram cloud communication
• GPS satellite communication

5. Non-Functional Requirements

5.1 Performance Requirements

• Response time should be less than 2 seconds.
• Obstacle detection should operate in real time.
• GPS updates should occur periodically.
• System should support continuous execution.

5.2 Reliability Requirements

• System should maintain stable performance.
• Emergency alerts should be highly reliable.
• Failure recovery mechanisms should be included.

5.3 Security Requirements

- User location data should be protected.
- Authentication should be implemented for caregivers.
- API keys and tokens should be secured.

5.4 Usability Requirements

- Interface should be simple and voice friendly.
- Audio feedback should be clear and understandable.
- Navigation instructions should be concise.

5.5 Maintainability Requirements

- Modular architecture should be used.
- AI models should support future updates.
- Code should follow proper documentation standards.

5.6 Scalability Requirements

- Additional sensors can be integrated in the future.
- Cloud support can be added later.
- Multi-user caregiver support should be possible.

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6. System Architecture

6.1 High-Level Architecture

The Dristi system consists of:

1. Input Layer

2. Camera

3. GPS

4. Voice Input

5. Processing Layer

6. Raspberry Pi

7. AI Models

8. Computer Vision Engine

9. Communication Layer
10. Telegram Bot API
11. Internet Services
12. Output Layer
13. Audio Feedback
14. Caregiver Notifications

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7. Use Case Descriptions

Use Case 1: Obstacle Detection

Item	Description
Actor	Visually Impaired User
Trigger	User starts navigation
Main Flow	Camera captures image → AI detects obstacle → Audio warning generated
Output	User receives spoken alert

Use Case 2: Emergency SOS

Item	Description
Actor	User
Trigger	Emergency button pressed
Main Flow	System obtains GPS → Sends Telegram alert
Output	Caregiver receives SOS notification

Use Case 3: Live Location Sharing

Item	Description
Actor	Caregiver
Trigger	Location request
Main Flow	GPS fetched → Telegram bot sends coordinates
Output	Live location displayed

8. Database Requirements

The system may store:

- User information
- Emergency contacts
- GPS history
- Activity logs
- System events

Possible databases:

- SQLite
- Firebase
- MongoDB

9. Future Enhancements

Potential future improvements include:

- Smart glasses integration
- Cloud-based analytics
- Indoor navigation support
- Face recognition
- Multi-language voice assistance
- Advanced deep learning models
- Mobile application integration
- Health monitoring sensors

10. Testing Requirements

10.1 Unit Testing

- Camera module testing
- GPS module testing
- Telegram bot testing
- Audio system testing

10.2 Integration Testing

- AI model integration
- Raspberry Pi integration
- End-to-end communication testing

10.3 Performance Testing

- Real-time detection latency

• GPS response timing
• CPU and memory utilization

10.4 User Acceptance Testing

• Navigation accuracy
• Voice response clarity
• Caregiver communication effectiveness

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11. Conclusion

Dristi is an AI-enabled assistive system aimed at improving the independence, mobility, and safety of visually impaired individuals. By integrating Computer Vision, IoT, GPS tracking, and voice interaction technologies, the system provides real-time assistance and caregiver connectivity.

The project has strong potential for future scalability, research contribution, and real-world social impact.